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ahmetkutuk1 Add p3d-intro vignette with usage examples ✓

f5cfa0a · 49 minutes ago

🕒 History

```
1  ---
2  title: "Introduction to p3d"
3  output: rmarkdown::html_vignette
4  vignette: >
5    %\VignetteIndexEntry{Introduction to p3d}
6    %\VignetteEngine{knitr::rmarkdown}
7    %\VignetteEncoding{UTF-8}
8  ---
9
10 ```{r setup, include = FALSE}
11 knitr::opts_chunk$set(
12   collapse = TRUE,
13   comment = "#>",
14   fig.width = 7,
15   fig.height = 5
16 )
17 library(p3d)
18 ```
19
20 ## Introduction
21
```

```

22 The p-value is a continuous measure of statistical evidence, yet in practice it is often reduced to a binary decision:  $p < 0.05$  is "significant
23
24 The p3d package visualizes the p-value of a single-sample t-test as a three-dimensional surface over two parameters:
25
26 - Effect size ( $d$ , Cohen's  $d$ )
27 - Sample size ( $n$ )
28
29 This vignette demonstrates the core functionality of the package.
30
31 ## The p-value surface
32
33 The central function of the package is p_surface(). Given a range of effect sizes and sample sizes, it computes the p-value for each combinatio
34
35 ```{r}
36 surface <- p_surface(
37   d = seq(-2, 2, 0.1),
38   n = seq(5, 200, 5)
39 )
40
41 str(surface, max.level = 1)
42 ```
43
44 The result is a list with several components:
45
46 - d: the vector of effect sizes
47 - n: the vector of sample sizes
48 - p: a matrix of p-values (dimensions length(d) × length(n))
49 - d_crit: critical effect sizes at  $\alpha = 0.05$  for each  $n$ 
50 - alpha: the significance level
51 - grid: a tidy data frame with all combinations
52
53 ### Inspecting the results
54
55 For example, the p-value at  $d = 0.5$ ,  $n = 30$  is:
56

```

```

57   ```{r}
58   surface$p[surface$d == 0.5, surface$n == 30]
59   ```
60
61   The critical effect size at  $n^* = 30$  is:
62
63   ```{r}
64   surface$d_crit[surface$n == 30]
65   ```
66
67   This means: with 30 observations, an effect size of at least 0.37 is required for statistical significance at  $\alpha = 0.05$ .
68
69   ## Static 3D visualization
70
71   The simplest way to visualize the surface is `plot_p_static()`, which uses base R graphics and has no external dependencies:
72
73   ```{r fig.alt = "Static 3D p-value surface"}
74   plot_p_static(surface)
75   ```
76
77   This is ideal for PDF output, journal figures, and printed reports.
78
79   ## Interactive 3D visualization
80
81   For interactive exploration, use `plot_p_surface()` with the `plotly` engine:
82
83   ```{r eval = FALSE}
84   plot_p_surface(surface, engine = "plotly")
85   ```
86
87   Alternatively, the `rgl` engine provides a native R window:
88
89   ```{r eval = FALSE}
90   plot_p_surface(surface, engine = "rgl")
91   ```

```

```

92
93 Both engines allow rotation, zooming, and panning.
94
95 ## 2D comparison
96
97 For comparison with existing 2D packages such as `pvaluefunctions`, use `p_curve_2d()` :
98
99 ```{r}
100 curve_30 <- p_curve_2d(d = seq(-2, 2, 0.1), n = 30)
101 head(curve_30)
102 ```
103
104 The 2D curve is mathematically equivalent to a slice of the 3D surface at fixed *n*.
105
106 ## Real-data example
107
108 Let's apply `p3d` to a real dataset. The `mtcars` dataset contains fuel consumption (mpg) for 32 automobiles. Suppose we want to test whether the
109
110 ```{r}
111 x <- mtcars$mpg
112 d_obs <- (mean(x) - 20) / sd(x)
113 n_obs <- length(x)
114
115 cat("Observed effect size:", round(d_obs, 4), "\n")
116 cat("Sample size:", n_obs, "\n")
117 ```
118
119 The p-value for this single observation is:
120
121 ```{r}
122 2 * (1 - pt(abs(d_obs) * sqrt(n_obs), df = n_obs - 1))
123 ```
124
125 By placing this observation on the 3D surface, we can see how the same effect size would behave at different sample sizes:
126

```

```
127 ```{r fig.alt = "Observed effect size on the 3D surface"}
128 surface_real <- p_surface(
129   d = seq(-0.5, 0.5, 0.01),
130   n = seq(5, 200, 5)
131 )
132
133 plot_p_static(surface_real, theta = 30, phi = 20)
134 ```
135
136 ## Conclusion
137
138 The p3d package provides a three-dimensional perspective on the p-value. By visualizing the p-value simultaneously across effect size and sam
139
140 1. The continuous nature of the p-value (it is not a binary switch)
141 2. The moderating effect of sample size on statistical significance
142 3. The interaction between effect size and sample size
143
144 This complements traditional 2D visualizations and helps weaken the dichotomous thinking that plagues statistical practice.
145
146 ## References
147
148 - Helske, J., Helske, S., & Cooper, M. (2021). Visualizing Hypothesis Tests in Practice. The R Journal.
149 - Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Lawrence Erlbaum Associates.
```